

SUMMARY INDICES IN TREATMENT EFFECT ESTIMATION: WHEN ARE THEY USEFUL?

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The paper examines the common practice of combining multiple outcomes into a summary index to estimate a causal effect. For a common class of IV-like estimators, such estimates represent a weighted average of component-level effects, yet the weights are implicit in the construction procedures and are never reported. These weights are derived and analyzed: they can be negative and unrestricted in magnitude. In a simple model, negative weights arise naturally for components of the summary index more strongly associated with the underlying concept. The asymptotic distribution of the estimates is established. Conventionally reported confidence intervals can be invalid because they do not account for data-dependent construction of the summary index, yet the corrections are likely to be small in real applications. Since none of the commonly used summary index procedures incorporate the sources of variability relevant to the estimator's variance, the resulting estimates do not generally exhibit the high power often attributed to them. Yet, in a simple RCT-like scenario with homoskedasticity and a small treatment effect, it is possible to unambiguously rank the summary indices studied here by the mean squared error of the difference-in-means estimates. Data from three recent empirical studies illustrate the theoretical results. Some simple summary indices can be useful for simplifying the reporting of results and should be interpreted as an "average effect" on the components. Beyond this, the results provide no compelling reason to use summary indices. For serious empirical conclusions, analyzing and reporting effects on individual components is recommended.

1. INTRODUCTION

How does one measure happiness, mental health, or women's empowerment in order to quantitatively assess the effects of policies on these important yet broad concepts? Empirical researchers often combine multiple related metrics into a *summary index*. Such summary indices of psychological well-being, women's empowerment, or political polarization have become common left-hand-side variables in empirical economic analyses. Positive estimated effects on such indices usually serve as arguments for a positive effect of the treatment. This article studies current empirical practices for constructing summary indices, analyzes the motivations behind their construction, and provides guidance on using summary indices for treatment effect evaluation.

There are several approaches to constructing summary indices. This paper focuses on a family of procedures characterized by the following steps. First, the components comprising a summary index are oriented in the same direction and converted to a common scale, usually through standardization using the control group. Then, a weighted sum of the standardized components is computed, and the summary index equals that sum, sometimes standardized once more. This general construction scheme gives rise to a host of procedures that differ in whether and how standardization is used, and in what weights are used in the second step. This family can be divided into two subfamilies based on how the weights are computed.

Summary indices constructed by averaging control-group-standardized components are referred to here as scale-normalized (**SN**) summary indices. This approach became popular following [Kling, Liebman, and Katz \(2007\)](#), who averaged control-group-standardized measures of depression, distress, sleep quality, and other metrics into a mental health summary index used to estimate the effect of moving to a safer neighborhood. Some recent studies following this summary index construction include: [Chetty et al. \(2011\)](#), [Casey, Glennerster, and Miguel \(2012\)](#), [Finkelstein et al. \(2012\)](#), [Banerjee et al. \(2015\)](#), [Hoynes, Schanzenbach, and Almond \(2016\)](#), [Blattman, Jamison, and Sheridan \(2017\)](#), [Bursztyn, González, and Yanagizawa-Drott \(2020\)](#), [Christensen et al. \(2021\)](#), [Levy \(2021\)](#), [Stantcheva \(2021\)](#), [Braghieri, Levy, and Makarin \(2022\)](#), [Bhatt et al. \(2024\)](#), and [Hawkins et al. \(2025\)](#).

In another approach, popularized by [Anderson \(2008\)](#), weights are determined by the inverse of a covariance matrix. These are referred to here as inverse-covariance (**IC**) summary indices. Recent studies following this summary index construction include: [Currie et al. \(2015\)](#), [Haushofer and Shapiro \(2016\)](#), [Cantoni et al. \(2017\)](#), [Cantoni et al. \(2019\)](#), [Chen and Yang \(2019\)](#), [Alfonsi et al. \(2020\)](#), [Allcott et al. \(2020\)](#), [Asher and Novosad \(2020\)](#), [Baranov et al.](#)

(2020), Allcott, Gentzkow, and Song (2022), Egger et al. (2022), Carranza et al. (2022), Beraja et al. (2023), Ramos-Toro (2023), Kinnan et al. (2024), and Baseler et al. (2025).

Causal effects on these summary indices are estimated using standard econometric techniques: linear regression, instrumental variables, difference-in-differences, and regression discontinuity designs. These estimators belong to the class of so-called IV-like estimators (Mogstad, Santos, and Torgovitsky (2018)), i.e., they can be represented as weighted sums of the outcomes with procedure-specific data-dependent weights. Therefore, for this class of estimators, owing to the linearity of summary index construction and estimation, *the treatment effect on a summary index equals a weighted sum of the treatment effects on its component outcomes*. These weights, however, are implicit in the summary index construction and are never reported; this article derives their explicit expressions. For **SN** summary indices, the weights are guaranteed to be positive and sum to one; for **IC** summary indices, the weights can be negative, their magnitude is unrestricted, and they do not sum to one. As a result, the interpretation of the estimated effect on an **IC** summary index can be problematic. It need not be positive (negative) even when the treatment positively (negatively) affects every component of the index, and it can artificially inflate the effect’s magnitude by placing excessive weight on some components. The paper considers a simple one-factor model in which each component of the summary index is a linear function of a latent scalar factor and uncorrelated noise. In this model, negative weights for the **IC** summary index arise naturally for components that are more strongly associated with the latent factor. Computing the weights for several recently published articles with available data confirms that such problematic weights arise in real datasets. Empirical researchers therefore should be careful when using such summary indices.

Some empirical articles motivate the use of summary indices by appealing to the improved precision of the resulting estimates, yet provide no formal results.³ This paper shows that such broad claims about efficiency gains from summary indices appear to lack theoretical support. None of the commonly used summary indices generally minimizes the estimator’s asymptotic variance. Tests based on these indices can exhibit high power against certain local alternatives

³ “[SN index] gives us **maximal power** to detect an effect on social outcomes, if such an effect is present.” Banerjee et al. (2015) (p. 49)

Baranov et al. (2020) refer to **IC** as “**the most efficient** weighted average of a set of outcomes” (p. 838)

“An index [SN] is useful for **increasing power** to the extent that all underlying components move in the same direction...”, Bhatt et al. (2024) (p. 20)

“In order to examine heterogeneous effects with **more statistical precision**, column 6 reports on an inverse covariance weighted index [IC index] of labor and spending”, Kinnan et al. (2024) (p. 279)

“[SN index] **improves statistical power** to detect effects that go in the same direction within a domain” Kling, Liebman, and Katz (2007) (p. 89)

that are unknown *ex ante* and unlikely to be empirically relevant in every case, but power can be low against alternatives in orthogonal directions. As a result, no summary index uniformly dominates in terms of power. Yet in a simple RCT-like scenario with homoskedasticity and a small treatment effect estimated by difference in means between the outcomes in the treated and the control groups, it is possible to unambiguously rank summary indices by their asymptotic mean squared error (MSE), with a version of the **IC** summary index achieving the lowest MSE. As an alternative to using summary indices, one can analyze the effects on the components of the index separately. However, the resulting tests must be corrected for multiple hypotheses. A simple but conservative way to do so is based on Bonferroni inequalities. The paper shows that tests based on summary indices are not guaranteed to have higher power than such conservative Bonferroni-corrected tests applied to individual components. Concrete examples from real datasets are provided in Section 4.

This paper derives the asymptotic distribution of estimates that use summary indices, provides a closed-form expression for the asymptotic variance, and shows that the variance estimators used by default in standard statistical packages are incorrect when summary indices are mechanically used as the left-hand-side variable in regressions. While most of the derivations cover the case of a scalar summary index, the approach generalizes straightforwardly to settings with multiple summary indices constructed from potentially overlapping sets of component outcomes – a common practice in empirical studies. This allows the construction of jointly valid confidence intervals for effects on multiple indices, offering a simple yet rigorous alternative to resampling-based methods such as [Westfall and Young \(1993\)](#), which are prevalent in practice but require extensive computations.

Overall, individual component analyses seem to be strictly more informative about the studied effects than summary-index-based analysis. Yet **SN** summary indices can provide a reasonable summary of component-level treatment effects, and the estimated effect should be interpreted as the average effect across the components. Beyond that, this paper finds no clear benefit to summary indices, and in particular **IC** summary indices: their weights may obscure the assessment of the treatment, and they offer no statistical power advantage over **SN** summary indices outside the simple RCT-like scenario with the difference-in-means estimator. Nonetheless, should a researcher use summary indices, it appears useful to explicitly report the weights applied to the treatment effects on the summary index’s components and to check that these weights are positive. The explicit expressions provided in this paper for **SN** and **IC** summary indices make doing so straightforward. Reporting the weights would make the analysis more transparent and give readers and decision makers more information about the drivers of the reported causal effects.

Literature This work relates to the literature on treatment evaluation with multiple outcomes, which dates back to Hotelling (1931), who introduced omnibus tests based on the T^2 statistic. Since such tests provide no information about the direction of departures from the null, their usefulness is limited for decision-making, as it is crucial to know whether the treatment benefits or harms recipients. To address this limitation, O'Brien (1984) proposed projection-based tests that target specific one-dimensional alternatives. The two statistics studied in O'Brien (1984) are linear combinations of sample mean differences across outcomes: one a simple average, the other weighted by the inverse variance-covariance matrix – the direct precursors of the **SN** and **IC** summary indices examined in this paper, respectively. Related tests for multiple endpoints in clinical trials are discussed by Pocock, Geller, and Tsiatis (1987). These ideas later entered applied economics through (i) Kling, Liebman, and Katz (2007), who introduced the **SN** index to evaluate the Moving to Opportunity program, and (ii) Anderson (2008), who proposed the **IC** index to study the effects of early intervention programs.

A related but distinct literature studies inference with multiple outcomes while retaining component-level hypotheses, typically through adjustments for multiple testing (e.g., Westfall and Young (1993), Romano and Wolf (2005), List, Shaikh, and Xu (2019)). That literature controls error rates across a family of tests; this paper instead studies the estimand, weights, and variance induced by constructing a summary index. Multiple testing becomes relevant below only as a benchmark for componentwise inference.

Independent and concurrent work by Lau (2026) is related, but the two papers differ in scope and exposition. Lau (2026) develops a decision-theoretic approach to choosing index weights, whereas this paper studies the properties of the existing **SN** and **IC** procedures when treatment effects are estimated using the class of IV-like estimators introduced in Section 2. The papers nevertheless obtain some common results, including that **IC** can assign negative weights. *Notation* Throughout the paper, standard Latin or Greek letters generally denote both vectors and scalars, without relying on boldface. Boldface is reserved for cases where it is important to indicate that an object is a vector or a matrix (rather than a scalar). For any integer k , $\vec{1}_k$ denotes the k -dimensional vector of ones, and $I_k = (e_{1,k} \dots e_{k,k})$ is the $k \times k$ identity matrix. All $o(\cdot)$ and $o_p(\cdot)$ terms are to be understood as referring to the limit $n \rightarrow \infty$, where, as usual, n denotes the sample size.

2. SETUP AND NOTATION

Data description and summary index construction The observable data, $\{W_i\}_{i=1}^n := \{\mathbf{y}_i, D_i, x_i\}_{i=1}^n$, consist of a vector of outcomes, $\mathbf{y}_i \in \mathbb{R}^p$; a cause (treatment, policy inter-

vention, etc.), $D_i \in \mathcal{D} \subseteq \mathbb{R}$, whose effect on $\mathbf{y}_i$ is of interest; and additional covariates and instruments, $\mathbf{x}_i \in \mathbb{R}^{k-1}$, which, for convenience, include a constant term as their first component. All independent variables are grouped into a single vector, $\tilde{\mathbf{x}}_i = (D_i, \mathbf{x}_i')' \in \mathbb{R}^k$.

The **SN** summary index, introduced by [Kling, Liebman, and Katz \(2007\)](#), is constructed as follows:

PROCEDURE 1: *Scale Normalized Summary Index.*

1. Switch signs in $\mathbf{y}$ if necessary so that the positive direction indicates a “better” outcome;
2. Demean all outcomes in $\mathbf{y}$ by subtracting the control-group mean and dividing each outcome by its control-group standard deviation, calling the transformed vector of outcomes $\tilde{\mathbf{y}}$;
3. Compute the summary index $\mathbf{s} := (s_1, \dots, s_n)'$ as:

$$s_i = (\tilde{\mathbf{1}}_p' \tilde{\mathbf{1}}_p)^{-1} \tilde{\mathbf{1}}_p' \tilde{\mathbf{y}}_i$$

4. Optionally, standardize $\mathbf{s}$ once more, i.e., repeat Step 2 for $\mathbf{s}$.

The **IC** summary index, introduced by [Anderson \(2008\)](#), is constructed as follows:

PROCEDURE 2: *Inverse Covariance Summary Index.*

1. Switch signs in $\mathbf{y}$ if necessary so that the positive direction indicates a “better” outcome;
2. Demean all outcomes⁴, divide each outcome by its control-group standard deviation, call the transformed vector of outcomes $\tilde{\mathbf{y}}$, and compute the variance-covariance matrix of $\tilde{\mathbf{y}}$, $\tilde{\mathbf{V}}_n := \widehat{\text{Var}}[\tilde{\mathbf{y}}]$;
3. Compute the summary index $\mathbf{s} := (s_1, \dots, s_n)'$ as:

$$s_i = (\tilde{\mathbf{1}}_p' \tilde{\mathbf{V}}_n^{-1} \tilde{\mathbf{1}}_p)^{-1} \tilde{\mathbf{1}}_p' \tilde{\mathbf{V}}_n^{-1} \tilde{\mathbf{y}}_i$$

4. Optionally, standardize $\mathbf{s}$ once more, i.e., repeat Step 2 for $\mathbf{s}$.

Unlike [Procedure 1](#), [Procedure 2](#) incorporates the inverse variance-covariance matrix of the outcomes. [Anderson \(2008\)](#) refers to this weighting scheme as Generalized Least Squares (GLS) weighting and motivates it as follows: “The GLS weighting procedure [...] increases efficiency

⁴In some studies, outcomes are demeaned by subtracting the control-group mean as in Procedure **SN**. As shown later, this difference is immaterial for the results (see the discussion in the beginning of Section 3.1).

by ensuring that outcomes that are highly correlated with each other receive less weight, while outcomes that are uncorrelated and thus represent new information receive more weight.” Yet the label “GLS” is somewhat misleading, since the GLS estimator uses the inverse of the *error-term* variance-covariance matrix, not that of the outcomes. However, some empirical works (e.g., [Bessone et al. \(2021\)](#)) do follow the true GLS approach by using the inverse covariance matrix of the estimated residuals. In what follows, I focus on Procedures 1 and 2; applied works often use modifications of these procedures: multiple summary indices are constructed from overlapping sets of outcomes or from other summary indices, and other summary index constructions exist based on principal components ([Filmer and Pritchett \(2001\)](#)) or economic weights ([Bhatt et al. \(2024\)](#)). See Appendix B for further discussion of how the results apply to these cases.

Estimand and estimator I consider the class of estimands that can be expressed as the expected value of $\mathbf{y}$ weighted by a mean-zero function of the data, $\omega(\cdot)$. [Mogstad, Santos, and Torgovitsky \(2018\)](#) refer to this class of estimands as *IV-like estimands*.

DEFINITION 1: IV-like estimand. An IV-like estimand induced by the weighting function $\omega(\cdot, \cdot)$ and a finite-dimensional parameter ν has the form:

$$\tau = E[\mathbf{y} \omega(\tilde{x}, \nu)] \in \mathbb{R}^p, \quad (1)$$

where $\omega(\tilde{x}, \nu)$ is a known (or identified) scalar-valued measurable function satisfying

$$E[\omega(\tilde{x}, \nu)] = 0. \quad (2)$$

As explained in [Mogstad, Santos, and Torgovitsky \(2018\)](#), least-squares (OLS), instrumental variables (IV), and two-stage least squares (TSLS) estimands fit this class; therefore, the setup covers estimates obtained in linear models, linear instrumental-variables models, two-way-fixed-effects models, difference-in-differences models, and (fuzzy) regression-discontinuity designs. Since the notation and dimensions of some variables do not perfectly coincide with those of [Mogstad, Santos, and Torgovitsky \(2018\)](#), I provide a table ([Table I](#)) of notable IV-like estimands and their corresponding weighting functions, similar to [Table II](#) in [Mogstad, Santos, and Torgovitsky \(2018\)](#).

TABLE I
NOTABLE IV-LIKE ESTIMANDS

Estimand	τ	$\omega(\tilde{x}, \nu)$	ν	Notes
Difference in means	$\bar{y}_1 - \bar{y}_0$	$(\nu(1 - \nu))^{-1}(D - \nu)$	$P(D = 1)$	–
OLS (jth compo- nents)	$E[\mathbf{y} \tilde{x}'] E[\tilde{x} \tilde{x}']^{-1} e_{j,k}$	$\tilde{x}' \nu e_{j,k}$	$E[\tilde{x} \tilde{x}']^{-1}$	–
IV (jth compo- nents)	$E[\mathbf{y} \tilde{z}'] E[\tilde{z} \tilde{x}']^{-1} e_{j,k}$	$\tilde{z}' \nu e_{j,k}$	$E[\tilde{z} \tilde{x}']^{-1}$	$\tilde{z} = (z, 1, x')'$, z-scalar
TSLs (jth compo- nents)	$E[\mathbf{y} \tilde{z}'] \Pi' (\Pi E[\tilde{z} \tilde{x}'])^{-1} e_{j,k}$	$\tilde{z}' \nu e_{j,k}$	$\Pi' (\Pi E[\tilde{z} \tilde{x}'])^{-1}$	$\Pi =$ $E[\tilde{x} \tilde{z}'] E[\tilde{z} \tilde{z}']^{-1}$, z-vector

Notes: The table presents some notable IV-like estimands with corresponding weighting functions $\omega(\cdot)$ and auxiliary parameters ν . Since $\mathbf{y}$ is p –dimensional vector, jth components of OLS, IV or TSLs estimates are also p -dimensional vectors.

The unknown population expectations and nuisance parameters (ν) are replaced by their sample analogues and estimates, respectively, giving rise to an estimator that, by analogy, can be referred to as an IV-like estimator.

DEFINITION 2: IV-like estimator. An IV-like estimator induced by the weighting function $\omega(\cdot, \cdot)$ and the finite-dimensional parameter ν has the form:

$$\hat{\tau}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i \omega(\tilde{x}_i, \hat{\nu}_n), \quad (3)$$

where $\hat{\nu}_n$ is an estimate of ν satisfying $\frac{1}{n} \sum_{i=1}^n \omega(\tilde{x}_i, \hat{\nu}_n) = 0$.

In practice, researchers often apply these estimators to summary indices in place of, or in addition to, applying them to the original outcomes $\mathbf{y}$. This yields a scalar estimator:

$$\hat{\beta}_n = \frac{1}{n} \sum_{i=1}^n s_{i,n} \omega(\tilde{x}_i, \hat{\nu}_n), \quad (4)$$

where $s_{i,n}$ denotes the summary index value of $\mathbf{y}_i$. In what follows, I sometimes refer to $\hat{\beta}_n$ as the *summary index estimator*.

3. THEORETICAL RESULTS

Section 3.1 demonstrates that the summary index estimator $\hat{\beta}_n$ equals a weighted sum of the components of $\hat{\tau}_n$, an IV-like estimator computed with the full outcome vector $\mathbf{y}$. The expressions for the weights are provided in Table II. Section 3.2 pertains to large sample statistical properties of $\hat{\beta}_n$, with Section 3.3 focus specifically on efficiency (mean squared error and statistical power of tests).

3.1. INTERPRETATION

PROPOSITION 1 *Let $\hat{\tau}_n$ be a linear IV-like estimator as in Definition 2, and $\hat{\beta}_n$ be the summary index estimator as in (4), where the summary index is computed by Procedure 1 or 2. Then*

$$\hat{\beta}_n = A_n \hat{\tau}_n,$$

for some row vector of weights A_n .

Thus, if $\hat{\beta}_n$ denotes the estimated treatment effect, then the treatment effect on a summary index equals a weighted sum of the treatment effects on its components⁵. The two-step argument behind this proposition is as follows. First, any summary index produced by Procedures 1 or 2 can be represented as:

$$s_{i,n} = \tilde{A}_n \mathbf{y}_i + b_n, \tag{5}$$

for some vector $\tilde{A}_n$ and scalar b_n . This follows from the fact that each step in Procedures 1 and 2 – subtraction, scalar or matrix multiplication, and addition – is an affine operation, and the successive application of affine transformations preserves the affine form in (5). Second, the linearity of the estimator (4) and the properties of $\hat{\nu}_n$ imply:

$$\hat{\beta}_n = \frac{1}{n} \left(\tilde{A}_n \sum_{i=1}^n \mathbf{y}_i \omega(\tilde{x}, \hat{\nu}_n) + b_n \sum_{i=1}^n \omega(\tilde{x}, \hat{\nu}_n) \right) = \tilde{A}_n \hat{\tau}_n,$$

hence the proposition, with $A_n = \tilde{A}_n$.

This result has two consequences. First, it is immaterial whether outcomes are demeaned and, if they are, whether demeaning is done using the control-group mean or the full-sample

⁵Kling, Liebman, and Katz (2007) make the same observation but only for the difference-in-means estimator and the SN summary index (see their footnote 10, p. 89). In fact, the relation holds much more broadly: it applies to the entire class of IV-like estimators and to any summary index that can be represented as an affine transformation.

mean. Any constant b_n cancels out by the properties of the estimator. Thus, the demeaning step in Procedures 1 and 2 can be skipped, as it does not affect the estimate. Second, if some components of $\mathbf{y}$ are missing, the currently common approach involves imputing the missing components. This implies that the weights assigned to the components of $\mathbf{y}$ when the summary index is constructed can differ across observations, depending on whether components are missing. This breaks the equality in Proposition 1. Further details on how missing data affect summary index construction and estimation are provided in Appendix C.

Thus, estimating the treatment effect on a summary index provides a simple yet indirect way to summarize the vector of component-level treatment effects in the form of their weighted sum. I next analyze the weights. Let $\hat{S}_0$ be a $p \times p$ diagonal matrix with the control-group standard deviations of $\mathbf{y}$ on its main diagonal. Using this matrix, define the vector of control-group-standardized treatment effects:

$$\hat{\tau}_{n, \text{st}} := \hat{S}_0^{-1} \hat{\tau}_n. \quad (6)$$

The transformation in (6) ensures that treatment effects are measured in units of their corresponding control-group standard deviations, so computing a weighted average of them is more meaningful. When all components of the summary index are measured on the same scale, this transformation appears to be unnecessary.

PROPOSITION 2 *Let $\hat{\tau}_{n, \text{st}}$ be the vector of control-group-standardized IV-like estimates and $\hat{\beta}_n$ be the summary index estimate. Then*

(i) *For **SN** summary index without final standardization (step 4 of Procedure 1):*

$$\hat{\beta}_n = \frac{1}{p} \mathbf{1}_p' \hat{\tau}_{n, \text{st}}$$

Thus, the summary index estimator is the arithmetic mean of the control-group-standardized component treatment effects.

(ii) *For **SN** summary index with final standardization:*

$$\hat{\beta}_n = c \frac{1}{p} \mathbf{1}_p' \hat{\tau}_{n, \text{st}},$$

for some nonzero $c \in \mathbb{R}$. Thus, the summary index estimator is a multiple of the arithmetic mean of the control-group-standardized component treatment effects.

(iii) *For **IC** summary index:*

$$\hat{\beta}_n = A_n^{\text{IC}} \hat{\tau}_{n,\text{st}},$$

where the entries of A_n^{IC} can exceed 1 in magnitude or be negative, and $A_n^{\text{IC}} \vec{1}_p$ need not equal 1.

The argument behind this proposition is as follows. First, [Table II](#) below provides expressions for the weights induced by the summary index procedures, which imply points (i) and (ii) of [Proposition 2](#). These weight expressions are derived in [Proposition 7](#) in Appendix A. Second, point (iii) of [Proposition 2](#) is illustrated using real datasets from recently published articles.

As [Proposition 2](#) implies, only a **SN** summary index without final standardization gives an adequate summary of the control-group-standardized effects on the components of the summary index, in the form of their arithmetic mean. If the index is standardized, the reported effect on the summary index artificially inflates or deflates that arithmetic mean, depending on whether c is larger than 1 or not. For an **IC** summary index, negative weights can give the summary index estimate $\hat{\beta}_n$ the opposite sign from $\hat{\tau}_{n,\text{st}}$'s components when those components all share the same sign. Thus, if the **IC** summary index is used, it is highly recommended to compute and report the weights that the procedure assigns to the components. The formulas for these weights are provided in [Table II](#).

TABLE II

WEIGHTS ASSIGNED TO COMPONENT TREATMENT EFFECTS BY DIFFERENT SUMMARY INDICES

Summary index	Weight vector
SN no final standardization	$p^{-1} \vec{1}_p'$
SN with final standardization	$(\vec{1}_p' \hat{S}_0^{-1} \hat{\Sigma}_0 \hat{S}_0^{-1} \vec{1}_p)^{-\frac{1}{2}} \vec{1}_p'$
IC no final standardization	$(\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p)^{-1} \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0$
IC with final standardization	$(\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{\Sigma}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p)^{-\frac{1}{2}} \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0$

Notes: The table reports the weights that each summary index assigns to the control-group-standardized component-level treatment effects, $\hat{\tau}_{n,\text{st}}$, when computing the summary index estimate, $\hat{\beta}_n$. The weight vector corresponds to $A_n \hat{S}_0$, where A_n denotes the weights assigned to the unstandardized estimates, $\hat{\tau}_n$, as defined in [Proposition 1](#). The matrices $\hat{\Sigma}$ and $\hat{\Sigma}_0$ denote the sample covariance matrices of $\mathbf{y}$ calculated using the full sample and the control-group subsample, respectively.

3.1.1. WHEN WEIGHTS ARE NEGATIVE?

Summary indices are sometimes motivated by the idea that several observed outcomes are proxies for the same latent concept. The following single-factor measurement model formalizes that idea:

$$\mathbf{y} = \lambda F + \varepsilon, \varepsilon \sim (0, \Psi), \quad (7)$$

where F is an unobserved, latent scalar factor, λ is p -dimensional vector of factor loadings.

ASSUMPTION 1:

1. Errors are uncorrelated: $\Psi = \text{Diag}(\sigma_1^2, \dots, \sigma_p^2)$.
2. The single factor is independent of measurement noise: $F \perp \varepsilon$.
3. Treatment/covariates only shift the factor, not the measurement noise: $\varepsilon \perp \tilde{x}$.
4. Components of $\mathbf{y}$ are non-negatively associated with the factor: $\lambda_j \geq 0, \forall j = 1, \dots, p$ and $\lambda \neq 0$.

This is a simplified version of the model studied by [Jöreskog and Goldberger \(1975\)](#).

PROPOSITION 3 *Consider the single-factor model (7) under [Assumption 1](#). Let $\text{Var}_0(y_j)$ denote the control-group variance of y_j , and $A^{\text{IC}} = (a_1, \dots, a_p)'$ be the population version of a vector of weights corresponding to **IC** summary index. Then there exists a threshold $r > 0$ such that*

$$a_j < 0 \Leftrightarrow \frac{\text{Cov}(y_j, F)}{\sqrt{\text{Var}_0(y_j)}} > r.$$

Thus, as [Proposition 3](#) implies, in the single-factor model, variables more strongly associated with the factor are more likely to receive a negative weight in **IC** summary index. The argument behind the proposition is based on applying the Sherman-Morrison formula to express Σ^{-1} . The details are in Appendix A.

3.2. LARGE SAMPLE STATISTICAL PROPERTIES OF ESTIMATES

I assume that the estimator admits the following representation.

ASSUMPTION 2: *Asymptotic linearity of estimates.*

$$\hat{\tau}_n = \tau + \frac{1}{n} \sum_{i=1}^n \vartheta(W_i) + o_p\left(\frac{1}{\sqrt{n}}\right), \quad (8)$$

for a measurable *influence function* $\vartheta(\cdot)$ such that $E[\vartheta(W)] = 0$, $E[\vartheta(W)\vartheta'(W)] < \infty$.

All notable IV-like estimators generally satisfy this assumption, and expressions for their influence functions ϑ are known.

EXAMPLE 1: *OLS Estimate.* For the linear projection:

$$\mathbf{y} = \tau D + \Gamma x + u, E[u \tilde{x}'] = 0, E_P[\tilde{x}\tilde{x}']^{-1} = \nu,$$

and OLS estimate:

$$\hat{\tau}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i \tilde{x}_i' \underbrace{\left(\frac{1}{n} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i' \right)^{-1}}_{:=\hat{\nu}} e_{1,k} = \quad (9)$$

$$= \tau + \frac{1}{n} \sum_{i=1}^n u_i \tilde{x}_i' \nu e_{1,k} + \frac{1}{n} \sum_{i=1}^n u_i \tilde{x}_i' (\hat{\nu}_n - \nu) e_{1,k}, \quad (10)$$

so in this case:

$$\vartheta(W; \tau, \Gamma, \nu) = u \tilde{x}' \nu e_{1,k} = (\mathbf{y} - [\tau, \Gamma] \tilde{x}) \tilde{x}' \nu e_{1,k}, \quad (11)$$

where, as before, $e_{1,k} = (1, 0, \dots, 0)'$. The last term in (9) can be shown to be $o_p\left(\frac{1}{\sqrt{n}}\right)$ under textbook least-squares regularity assumptions

The representation in [Assumption 2](#) can be obtained generically by computing the Hadamard derivative of an IV-like estimand with respect to the data distribution. This approach has been used to show that the weights that summary indices assign to the components admit a similar representation.

ASSUMPTION 3: *Asymptotic linearity of weights.*

$$\hat{A}_n = A + \frac{1}{n} \sum_{i=1}^n \psi(W_i) + o_p\left(\frac{1}{\sqrt{n}}\right), \quad (12)$$

for some non-stochastic A and a measurable function $\psi(\cdot)$ such that $E[\psi(W)] = 0$, $E[\psi(W)\psi'(W)] < \infty$.

[Proposition 8](#) in Appendix A provides an expression for the influence function ψ for the non-standardized SN summary index – the only procedure among studied here that guarantees an adequate interpretation of the treatment effect estimate.

PROPOSITION 4 – Large sample properties of $\hat{\beta}_n$: Under [Assumptions 2 and 3](#), $\hat{\beta}_n$ is consistent for $\beta := A\tau$,

$$\hat{\beta}_n \xrightarrow{p} \beta \text{ as } n \rightarrow \infty,$$

and asymptotically normal,

$$\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \varsigma) \text{ as } n \rightarrow \infty, \quad (13)$$

where

$$\varsigma = \text{Var}(A \vartheta(W) + \psi(W) \tau), \quad (14)$$

and $\vartheta(W) \in \mathbb{R}^{p \times 1}$, $\psi(W) \in \mathbb{R}^{q \times p}$ are the influence functions defined in [Assumptions 2 and 3](#).

The proposition follows from taking the product of the two representations in [Assumptions 2 and 3](#):

$$\hat{\beta}_n = \underbrace{A\tau}_{=\beta} + \frac{1}{n} \sum_{i=1}^n (A \vartheta(W_i) + \psi(W_i) \tau) + o_p\left(\frac{1}{\sqrt{n}}\right),$$

and then applying the Lindeberg-Levy central limit theorem. A short proof is included in Appendix A for completeness.

Note that when summary indices are used as left-hand-side variables in regressions, commonly used statistical packages produce standard errors that generally fail to estimate the true variability of $\hat{\beta}_n$ as given by the asymptotic variance expression in [\(14\)](#).

EXAMPLE 1 (CONTINUED): *OLS Estimate.* Let $(\hat{\beta}_n, \hat{\Lambda}_n)$ be OLS estimates from:

$$s = \hat{\beta}_n D + \hat{\Lambda}_n x + \hat{\varepsilon}, \quad (15)$$

and $(\hat{\tau}_n, \hat{\Gamma}_n)$ be OLS estimates from:

$$y = \hat{\tau}_n D + \hat{\Gamma}_n x + \hat{u}.$$

One can show that $\hat{\varepsilon}_i = A_n \hat{u}_i, \forall i$. The reported sandwich variance estimate from regression (15) (for example, the HC1 version) then equals:

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta}_n) &= e'_{1,k} \left(\left(\frac{1}{n} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i' \hat{\varepsilon}_i^2 \right) \left(\frac{1}{n} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i' \right)^{-1} \right) e_{1,k} = \\ &= \frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i (e'_{1,k} \hat{\nu}_n \tilde{x}_i) (\tilde{x}_i' \hat{\nu}_n e_{1,k}) \hat{\varepsilon}_i = A_n \left(\frac{1}{n} \sum_{i=1}^n \hat{u}_i (e'_{1,k} \hat{\nu}_n \tilde{x}_i) (\tilde{x}_i' \hat{\nu}_n e_{1,k}) \hat{u}_i' \right) A_n', \end{aligned}$$

which, because of (11), at best estimates $\text{Var}(A \vartheta(W))$ and ignores the second term in (14), arising from the variability of the weights. Generally this leads to invalid confidence intervals.

Therefore, because conventional packages mechanically ignore the variability in summary index weights, the reported confidence intervals for β are generally invalid. However, if weights were in fact non-stochastic and hence $\psi \equiv 0$, or if $\tau = 0$, the term $\psi(W) \tau$ in the variance expression (14) equals zero, and so the conventional packages would estimate the variance correctly. This further implies that (i) the t-test reported in the summary index regression has the correct asymptotic rate of type I errors (size) if the null hypothesis is $\tau = 0$, and (ii) if $\tau \approx 0$, the correction to the asymptotic variance is small and unlikely to substantially alter conclusions about the significance of the estimates, as confirmed in some of the empirical examples presented above.

3.3. EFFICIENCY

Several empirical studies motivate summary indices by appealing to gains in efficiency or statistical power (see e.g., Footnote 3). The results above show that such claims do not hold in general. In (14), the asymptotic variance of $\hat{\beta}_n$ equals $\text{Var}(A \vartheta(W) + \psi(W) \tau)$; hence it

depends on the true treatment-effect vector τ , the limiting weights A , and the influence functions of both the component estimator and the estimated weights. Therefore, neither **SN** nor **IC** can be uniformly most efficient without further restrictions.

I next consider local alternatives $\tau_n = \frac{\delta}{\sqrt{n}}$. This sequence represents alternatives whose magnitude shrinks at the same rate as the sampling error of $\hat{\tau}_n$ and is commonly used to compare the asymptotic power of tests (see e.g., [van der Vaart \(2000\)](#)). Under these alternatives, the contribution of the estimated weights to the asymptotic variance vanishes. I derive the limiting local mean squared error (MSE) of $\hat{\beta}_n$ and the local power of the corresponding two-sided t -test, and characterize the treatment-effect direction against which a given summary index test has maximal local power. I then show that summary index procedures cannot generally be ranked by local MSE, although such a ranking is possible in a simple RCT-like scenario with homoskedasticity where the treatment effect is estimated by difference in means. Finally, I derive the local power of a conservative Bonferroni-corrected test applied to the individual components. This power may exceed the local power of summary-index-based tests; examples in which the Bonferroni test has higher local power are provided in Section 4.

ASSUMPTION 4: Local DGP. Consider the sequence of local alternatives $\tau_n = \frac{\delta}{\sqrt{n}}$, so that the estimator admits the representation

$$\hat{\tau}_n = \frac{\delta}{\sqrt{n}} + \frac{1}{n} \sum_{i=1}^n \vartheta(W_i) + o_p\left(\frac{1}{\sqrt{n}}\right), \quad (16)$$

where, under the n th local DGP, $E_n[\vartheta(W)] = 0$, $\text{Var}_n[\vartheta(W)] = V$ – a positive definite matrix – and $\delta \in \mathbb{R}^p \setminus \{0\}$ is the local alternative direction.

The scaling $\tau_n = \frac{\delta}{\sqrt{n}}$ captures the idea that τ is small in magnitude relative to the sample size. Consider the two-sided test,

$$\varphi_n = 1\{|T_n| > z_{1-\alpha/2}\}, \quad (17)$$

of $H_0 : \tau = 0$ that rejects when the t -statistic

$$T_n := \frac{\hat{\beta}_n}{\text{s.e.}(\hat{\beta}_n)}$$

exceeds, in absolute value, the $1 - \alpha/2$ standard normal quantile, $z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$. The asymptotic local power of the test is the limit along such a sequence of local alternatives of the probability of rejecting the null hypothesis,

$$\pi := \lim_{n \rightarrow \infty} P_n(\varphi_n = 1).$$

PROPOSITION 5 — Asymptotic local MSE, local power, and optimal direction: Under [Assumption 3](#) and [Assumption 4](#), for a scalar summary index with population weighting row vector A , the following statements hold.

(i) **(Local MSE)** If the mean-square regularity condition ([Assumption 6](#) in Appendix A) also holds, let $\beta_n := A\tau_n$ denote the local sequence of the population treatment effect on the summary index. The centered local mean squared error satisfies

$$\lim_{n \rightarrow \infty} nE_n \left[(\hat{\beta}_n - \beta_n)^2 \right] = AVA'. \quad (18)$$

(ii) **(Local power)** If $\sqrt{n} \text{s.e.}(\hat{\beta}_n) \xrightarrow{p} \sqrt{AVA'}$, then the asymptotic local power of the test in [\(17\)](#) equals

$$\pi(\Delta) := \Phi(\Delta - z_{1-\alpha/2}) + \Phi(-\Delta - z_{1-\alpha/2}), \quad (19)$$

where

$$\Delta := \frac{A\delta}{\sqrt{AVA'}}.$$

(iii) **(Optimal direction)** Among all local alternative directions δ with fixed $\delta'V^{-1}\delta$, $\pi(\Delta)$ is maximized at

$$\delta_A^* \propto VA'. \quad (20)$$

The argument behind (i) and (ii) is based on the fact that, under [Assumption 4](#),

$$\sqrt{n}(\hat{\beta}_n - \beta_n) \xrightarrow{d} \mathcal{N}(0, AVA'), \quad T_n \xrightarrow{d} \mathcal{N}(\Delta, 1).$$

The convergence in distribution above gives the local power statement; the mean-square regularity condition ([Assumption 6](#) in Appendix A) strengthens it to the second-moment convergence used for [\(18\)](#).

For part (iii), the Cauchy-Schwarz inequality applied to $V^{1/2}A'$ and $V^{-1/2}\delta$ gives

$$|A\delta|^2 = \left| (V^{1/2}A')' (V^{-1/2}\delta) \right|^2 \leq (AVA')(\delta'V^{-1}\delta),$$

with equality iff $V^{-1/2}\delta \propto V^{1/2}A'$, i.e., $\delta \propto VA'$. A formal proof is included in Appendix A.

COROLLARY 3.1 — Optimal directions for **SN** and **IC**: Suppose that, along the sequence of DGPs satisfying [Assumption 4](#), the sample covariance matrix of $\mathbf{y}$, $\hat{\Sigma}$, and the control-group covariance matrix, $\hat{\Sigma}_0$, satisfy $\hat{\Sigma} \xrightarrow{p} \Sigma$ and $\hat{\Sigma}_0 \xrightarrow{p} \Sigma_0$ as $n \rightarrow \infty$, where Σ and Σ_0 are positive definite, non-stochastic matrices. Under [Assumption 3](#), the optimal local directions in (20) for **SN** and **IC** summary indices are

$$\delta_{\text{SN}}^* \propto V S_0^{-1} \vec{1}, \quad \delta_{\text{IC}}^* \propto V \Sigma^{-1} S_0 \vec{1},$$

where $S_0 = \text{Diag}(\Sigma_0)^{\frac{1}{2}}$.

That is, among all local alternatives, the test (17) based on the **SN** (respectively the **IC**) summary index attains its highest asymptotic local power against treatment-effect directions proportional to δ_{SN}^* (respectively δ_{IC}^*). Both directions depend on unknown quantities and represent somewhat arbitrary directions in the space of possible treatment-effect directions. In one simple scenario, the results can be refined further and local MSEs can be ranked for the summary indices studied here.

ASSUMPTION 5: RCT with homoskedasticity, small effect, and difference-in-means. Let $(D_i, \mathbf{u}_i)$ be i.i.d., with $D_i \in \{0, 1\}$, $\nu := P(D_i = 1) \in (0, 1)$, $E_P[\mathbf{u}_i \mid D_i] = \mathbf{0}$, and $E_P[\mathbf{u}_i \mathbf{u}_i' \mid D_i] = \mathcal{V}$, where $\mathcal{V}$ is positive definite. The local outcomes satisfy

$$\mathbf{y}_{i,n} = \alpha + \frac{\delta}{\sqrt{n}} D_i + \mathbf{u}_i, i = 1, \dots, n.$$

The component-level treatment-effect estimator is the difference in means,

$$\hat{\tau}_n = \bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_0.$$

The sequence of data-generating processes satisfying [Assumption 5](#) can describe an RCT in which treatment D is randomly assigned, the variances of the error term are the same in the treated and control groups, the treatment effect is local, and component effects are estimated by difference in means.

PROPOSITION 6 — Local MSE ranking under difference-in-means: For each summary index procedure in [Table II](#), define MSE_{SN}^0 , MSE_{SN}^1 , MSE_{IC}^0 , and MSE_{IC}^1 as the limiting local MSEs in (18), computed from the difference-in-means estimator in [Assumption 5](#). Superscript 0 denotes a summary index procedure with no final standardization, and

superscript 1 denotes one with final standardization. Then, under [Assumption 5](#) and [Assumption 6](#),

$$\text{MSE}_{\text{IC}}^0 \leq \text{MSE}_{\text{SN}}^0 \leq \text{MSE}_{\text{SN}}^1 = \text{MSE}_{\text{IC}}^1 .$$

The argument behind this proposition relies on the fact that, under [Assumption 5](#), the difference-in-means estimator has asymptotic variance proportional to $\mathcal{V}$, while both $\hat{\Sigma}$ and $\hat{\Sigma}_0$ converge to $\mathcal{V}$. Plugging these quantities into [\(18\)](#) reduces the ranking to two applications of the Cauchy-Schwarz inequality. A formal proof is included in Appendix A.

As an alternative to constructing summary indices, one may estimate treatment effects on the components of the summary index and test the component null hypotheses individually, correcting for multiple testing. Let $T_{j,n} := \frac{\hat{\tau}_{j,n}}{\text{s.e.}}(\hat{\tau}_{j,n})$ be the component-level t -statistic and let $c_p := z_{1-\alpha/(2p)}$. The two-sided Bonferroni rule rejects $H_0 : \tau = \mathbf{0}$ whenever

$$\max_{1 \leq j \leq p} |T_{j,n}| > c_p .$$

The local power of this rule can be derived under [Assumption 4](#); the formal statement is given in [Corollary 1.1](#) in Appendix A. This power need not be below the local power of summary-index-based tests, and [Section 4](#) provides examples in which it is higher.

4. EMPIRICAL ILLUSTRATIONS

This section presents three recently published empirical studies that use summary indices. The goal is threefold: first, to illustrate how summary indices are constructed and used in practice; second, to exploit the availability of data in all three cases to compute the implicit weights that summary indices assign to the components ([Table II](#)), demonstrating that the problems outlined in [Proposition 2](#) are not merely theoretical possibilities but can arise in real datasets; third, for applications in which component-level estimates and covariance matrices can be reconstructed, to evaluate the local-power formulas derived in [Section 3.3](#) and compare summary-index-based tests with Bonferroni-corrected component tests.

4.1. EXAMPLE A: EFFECT OF FACEBOOK DEACTIVATION ([ALLCOTT ET AL. \(2020\)](#))

The article studies the effects of Facebook on (i) time allocation and substitution toward other online and offline activities, (ii) news knowledge and political polarization, (iii) subjective well-being, and (iv) post-experiment demand for and valuation of Facebook. It employs a variant of the IC summary index with standardization to quantify news knowledge, political polarization, and subjective well-being⁶. The paper’s main conclusions—that Facebook deactivation

causally reduces political polarization and increases subjective well-being—were supported by linear instrumental-variable estimates in which random assignment of the incentive to deactivate a Facebook account is used as an instrument for the endogenous decision to deactivate, and the aforementioned indices are used as outcomes.

The subjective well-being index combines survey-based measures of happiness, life satisfaction, loneliness, depression, anxiety, absorption, and boredom over the previous four weeks. It also includes momentary measures collected through daily text messages: current happiness, positive emotion, and the absence of loneliness. [Table III](#) reports these components and the corresponding weights that different summary index procedures assign to the component-level treatment effects, computed using the formulas in [Table II](#). The last row of [Table III](#) presents the sum of the weights. As the table shows, “happiness” receives a negative weight in the estimated treatment effect on the **IC** summary index of well-being. In addition, when the construction of the summary index includes final standardization, the weights sum to more than one.

The political polarization index combines measures of affective polarization toward the two major parties and the president, anger toward the opposing party, and exposure to news congenial to one’s own party. It also includes measures of polarization in political opinions, beliefs about current events, and voting preferences. [Table IV](#) reports the weights assigned to the components of this summary index. Similar problems are evident: the **IC** summary index assigns negative weights to some components, while procedures involving final standardization produce weights that sum to more than one.

⁶Instead of the covariance matrix, the paper computes the average inner product of control-group-standardized outcome vectors, but averaged over the entire sample (thus it is also not exactly the control-group covariance matrix), so the weights do not perfectly match those obtained from [Procedure 2](#), yet the differences are immaterial.

TABLE III

WELL-BEING SUMMARY INDEX: COMPONENTS AND WEIGHTS

Component	SN no std	SN std	IC no std	IC std
Happiness	0.10	0.14	-0.10	-0.15
Life satisfaction	0.10	0.14	0.08	0.12
Loneliness x (-1)	0.10	0.14	0.04	0.06
Depressed x (-1)	0.10	0.14	0.04	0.06
Anxious x (-1)	0.10	0.14	0.23	0.34
Absorbed	0.10	0.14	0.18	0.28
Bored x (-1)	0.10	0.14	0.24	0.37
SMS happiness	0.10	0.14	0.10	0.15
SMS positive emotion	0.10	0.14	0.11	0.17
SMS not lonely	0.10	0.14	0.08	0.12
Total sum	1.00	1.41	1.00	1.52

Notes: The table lists the components of the “Subjective well-being summary index” defined in [Allcott, Gentzkow, and Song \(2022\)](#) and, for each of the four summary-index procedures studied, the implicit component weights computed as in [Table II](#). The final row reports the total weight for each summary index. Boldface indicates negative weights and total weights greater than 1.

TABLE IV

POLITICAL POLARIZATION SUMMARY INDEX: COMPONENTS AND WEIGHTS

Component	SN no std	SN std	IC no std	IC std
Party affective polarization	0.14	0.21	-0.09	-0.15
Trump affective polarization	0.14	0.21	0.13	0.20
Party anger	0.14	0.21	0.20	0.32
Congenial news exposure	0.14	0.21	0.27	0.43
Issue polarization	0.14	0.21	0.15	0.24
Belief polarization	0.14	0.21	0.24	0.38
Vote polarization	0.14	0.21	0.11	0.17
Total sum	1.00	1.49	1.00	1.60

Notes: The table lists the components of the “Political polarization summary index” defined in [Allcott, Gentzkow, and Song \(2022\)](#) and, for each of the four summary-index procedures studied, the implicit component weights computed as in [Table II](#). The final row reports the total weight for each summary index. Boldface indicates negative weights and total weights greater than 1.

4.2. EXAMPLE B: EFFECT OF COGNITIVE BEHAVIORAL THERAPY ([BARANOV ET AL. \(2020\)](#))

The article studies the effects of cognitive behavioral therapy on women’s mental health, financial empowerment and decision-making, parental investment in children, and child development. It analyzes a cluster-randomized controlled trial (the Thinking Healthy Program) in rural Punjab, Pakistan, in which 903 women clinically diagnosed with prenatal depression were randomized to receive cognitive behavioral therapy. The article constructs multiple summary indices (depression severity index, financial empowerment index, parental investment summary index, etc.) using a variant of the **IC** procedure⁷. Using least-squares estimates with summary indices as left-hand-side variables, the study claims positive and persistent effects of the therapy on mental health and other areas of women’s life, such as financial empowerment and parental investments.

The depression severity summary index at the 1-year follow-up was computed from a depression indicator, the depression severity (Hamilton) score, the BDQ disability score, and the Global Assessment of Functioning (GAF) score, where the latter component receives a negative weight, as reported in [Table V](#). For the depression severity index at the 6-month follow-up, the **IC** summary index as described in [Procedure 2](#) does not produce a negative weight. However, due to the way the paper treats missing values when computing the summary index, their procedure does produce a negative weight for the depression severity (HAM-D) component.

The study also constructs a depression trajectory index from depression severity measured at the 6-month, 1-year, and 7-year follow-ups. As reported in [Table VI](#), some components of the 6-month depression severity index that previously did not receive negative weights begin to do so once included in this trajectory index.

⁷Because of the way the paper treats missing values, the implicit weights their procedure assigns to components are not equal to those resulting from [Procedure 2](#).

TABLE V

DEPRESSION SEVERITY INDEX AT 1 YEAR FOLLOW-UP: COMPONENTS AND WEIGHTS

Component	SN no std	SN std	IC no std	IC std
Depressed 1y	0.25	0.26	0.27	0.28
Depression severity 1y (HAM-D)	0.25	0.26	0.21	0.22
BDQ disability score 1y	0.25	0.26	0.60	0.63
General functioning 1y (GAF)	0.25	0.26	-0.09	-0.09
Total sum	1.00	1.05	1.00	1.04

Notes: This table reports the weights assigned to each depression trajectory component by four summary index procedures, following Baranov et al. (2020). **SN** no std and **SN** std are scale-normalized (equal-weight) procedures without and with final index standardization, respectively. **IC** no std and **IC** std are inverse-covariance-weighted procedures without and with final index standardization.

TABLE VI

MATERNAL DEPRESSION TRAJECTORY INDEX: COMPONENTS AND WEIGHTS

Component	SN no std	SN std	IC no std	IC std
Depressed (7y)	0.08	0.12	0.12	0.20
# Depression symptoms present (7y)	0.08	0.12	0.01	0.01
Symptoms cause impairment (7y)	0.08	0.12	0.14	0.23
Depressed in previous 2 years (7y)	0.08	0.12	0.32	0.53
Depressed (1y)	0.08	0.12	0.12	0.20
Depression severity 1y (Hamilton score)	0.08	0.12	0.07	0.12
BDQ disability score 1y	0.08	0.12	0.09	0.16
General functioning (GAF) 1y	0.08	0.12	-0.11	-0.19
Depressed 6m	0.08	0.12	0.14	0.23
Depression severity 6m (Hamilton score)	0.08	0.12	-0.01	-0.02
BDQ disability score 6m	0.08	0.12	0.16	0.26
General functioning (GAF) 6m	0.08	0.12	-0.05	-0.09
Total sum	1.00	1.47	1.00	1.64

Notes: This table reports the weights assigned to each depression trajectory component by four summary index procedures, following Baranov et al. (2020). **SN** no std and **SN** std are scale-normalized (equal-weight) procedures without and with final index standardization, respectively. **IC** no std and **IC** std are inverse-covariance-weighted procedures without and with final index standardization.

4.3. EXAMPLE C: EFFECT OF SKILL CERTIFICATION (CARRANZA ET AL. (2022))

The article studies whether providing additional information about workseekers' skills – and specifically, which side of the market receives it and how credibly it can be shared – affects job search behavior, firms' hiring decisions, and workseekers' employment and earnings. The study examines 6,891 young, predominantly Black African job seekers in metropolitan Johannesburg who had completed secondary school but had limited postsecondary education, work experience, and professional networks. A series of field experiments varied whether assessments of participants' skills results were disclosed privately to participants, credibly shared with employers, or provided only to firms. I refer to these treatment arms as private certification, public certification, and placebo certification, respectively, following the labels used in the replication files.

The researchers collect several employment-related outcomes, including earnings measures, labor-contract types, employment status at different points in time, and hours worked. These outcomes are used to construct several employment indices, some focusing on earnings and others on employment status and time spent working, as well as a broad labor market summary index that aggregates metrics across all three dimensions. All summary indices are constructed using the IC procedure. The components of these indices and their corresponding weights are reported in Table VII and Table VIII.

TABLE VII
ENDLINE EMPLOYMENT INDEX: COMPONENTS AND WEIGHTS

Component	SN no std	SN std	IC no std	IC std
Earnings	0.33	0.38	-0.20	-0.23
Wage	0.33	0.38	0.65	0.77
Formal work	0.33	0.38	0.54	0.64
Total sum	1.00	1.15	1.00	1.18

Notes: The table lists the components of the "Employment summary index" defined in Carranza et al. (2022) and, for each of the four summary-index procedures studied, the implicit component weights computed as in Table II. The final row reports the total weight for each summary index. Boldface indicates negative weights and total weights greater than 1.

TABLE VIII

LABOR MARKET INDEX: COMPONENTS AND WEIGHTS

Component	SN no std	SN std	IC no std	IC std
Employed	0.20	0.23	-0.81	-1.04
Hours	0.20	0.23	1.84	2.36
Earnings	0.20	0.23	-2.58	-3.30
Hourly wage	0.20	0.23	2.29	2.93
Written contract	0.20	0.23	0.26	0.33
Total sum	1.00	1.14	1.00	1.28

Notes: The table lists the components of the “Labor market summary index” defined in Carranza et al. (2022) and, for each of the four summary-index procedures studied, the implicit component weights computed as in Table II. The final row reports the total weight for each summary index. Boldface indicates negative weights and total weights greater than 1.

4.4. LOCAL POWER COMPARISON

Table IX reports plug-in local powers computed from empirical treatment-effect estimates and covariance matrices. These calculations use sample analogues of the population objects in Proposition 5 and Corollary 1.1, and should be read as numerical illustrations of the formulas rather than as estimates of exact finite-sample power in these applications. In the Facebook-deactivation and skill-certification applications, I use the estimated treatment-effect vectors without rescaling. In the maternal-depression application, the unscaled estimates are large relative to their standard errors, yielding local powers essentially equal to one. I therefore divide the estimated component treatment-effect vector by four before computing power; this preserves the estimated direction and reduces only its magnitude.

TABLE IX
LOCAL-POWER COMPARISONS USING EMPIRICAL ESTIMATES

Application	Index	SN	IC	Bonferroni	Ranking
Allcott et al.	Subjective well-being	0.563	0.597	0.704	Bonferroni > IC > SN
Allcott et al.	Political polarization	0.704	0.929	0.949	Bonferroni > IC > SN
Baranov et al.	Depression severity 1y	0.435	0.415	0.334	SN > IC > Bonferroni
Baranov et al.	Depression trajectory	0.579	0.513	0.387	SN > IC > Bonferroni
Carranza et al.	Employment index, public certification	0.984	0.950	0.988	Bonferroni > SN > IC
Carranza et al.	Employment index, private certification	0.608	0.557	0.520	SN > IC > Bonferroni
Carranza et al.	Labor market index, public certification	0.989	0.913	0.990	Bonferroni > SN > IC
Carranza et al.	Labor market index, placebo certification	0.078	0.062	0.049	SN > IC > Bonferroni

Notes: The table reports plug-in local power at $\alpha = 0.05$. For the maternal-depression application, the empirical component treatment-effect vector is divided by four before applying the local-power formula. All other applications use the empirical estimates without rescaling.

5. CONCLUSION

In conclusion, out of the summary indices studied here, a **SN** summary index without final standardization is useful for treatment effect evaluations using IV-like estimators: estimating the effect on it provides a fast way to compute and communicate the average of the control-group-standardized effects on its components. Beyond that, this paper finds no additional benefit to using summary indices, and therefore recommends relying on component-level empirical analyses for serious conclusions.

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A. PROOFS

PROPOSITION 7 — Derivation of the weights in [Table II](#): Let $\hat{S}_0$ be the diagonal matrix of control-group sample standard deviations, let $\hat{\Sigma}$ and $\hat{\Sigma}_0$ be the full-sample and control-group sample covariance matrices of $\mathbf{y}$, and define $\hat{\tau}_{n,\text{st}} := \hat{S}_0^{-1} \hat{\tau}_n$. For the four summary-index procedures in [Table II](#), the weight vectors on $\hat{\tau}_{n,\text{st}}$ are those reported in [Table II](#).

PROOF As discussed in [Section 3.1](#), demeaning in Step 2 only changes the constant term in the affine representation of the summary index. This constant cancels from the estimator because $\frac{1}{n} \sum_{i=1}^n \omega(\tilde{x}_i, \hat{\nu}_n) = 0$. Hence the proof can ignore demeaning and derive only the row vector A_n such that $s_i = A_n \mathbf{y}_i + b_n$; the weight vector in [Table II](#) is then $A_n \hat{S}_0$, since $\hat{\tau}_n = \hat{S}_0 \hat{\tau}_{n,\text{st}}$.

For the **SN** procedure without final standardization,

$$s_i = p^{-1} \vec{1}_p' \hat{S}_0^{-1} \mathbf{y}_i + b_n, A_n^{\text{SN}} = p^{-1} \vec{1}_p' \hat{S}_0^{-1}.$$

Therefore $A_n^{\text{SN}} \hat{S}_0 = p^{-1} \vec{1}_p'$, which gives the first row of [Table II](#).

If the **SN** index is standardized one more time, the control-group variance of the unstandardized index equals

$$p^{-2} \vec{1}_p' \hat{S}_0^{-1} \hat{\Sigma}_0 \hat{S}_0^{-1} \vec{1}_p.$$

Dividing by its square root gives

$$A_n^{\text{SN},\text{st}} = \left(\vec{1}_p' \hat{S}_0^{-1} \hat{\Sigma}_0 \hat{S}_0^{-1} \vec{1}_p \right)^{-\frac{1}{2}} \vec{1}_p' \hat{S}_0^{-1},$$

and hence

$$A_n^{\text{SN},\text{st}} \hat{S}_0 = \left(\vec{1}_p' \hat{S}_0^{-1} \hat{\Sigma}_0 \hat{S}_0^{-1} \vec{1}_p \right)^{-\frac{1}{2}} \vec{1}_p',$$

which is the second row.

For the **IC** procedure, let $\tilde{\mathbf{y}}_i = \hat{S}_0^{-1} \mathbf{y}_i$ after dropping irrelevant demeaning constants. The covariance matrix of $\tilde{\mathbf{y}}$ is

$$\tilde{V}_n = \hat{S}_0^{-1} \hat{\Sigma} \hat{S}_0^{-1}, \tilde{V}_n^{-1} = \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0.$$

Therefore, without final standardization,

$$s_i = \left(\vec{1}_p' \tilde{V}_n^{-1} \vec{1}_p \right)^{-1} \vec{1}_p' \tilde{V}_n^{-1} \tilde{\mathbf{y}}_i = A_n^{\text{IC}} \mathbf{y}_i + b_n,$$

where

$$A_n^{\text{IC}} = \left(\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p \right)^{-1} \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1}.$$

Multiplying by $\hat{S}_0$ gives the third row of [Table II](#):

$$A_n^{\text{IC}} \hat{S}_0 = \left(\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p \right)^{-1} \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0.$$

Finally, if the **IC** index is standardized one more time, the scalar normalization from the previous display cancels from numerator and denominator. The control-group variance of the unnormalized linear combination $\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \mathbf{y}_i$ is

$$\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p.$$

Thus

$$A_n^{\text{IC}, \text{st}} = \left(\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p \right)^{-\frac{1}{2}} \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1},$$

and hence

$$A_n^{\text{IC}, \text{st}} \hat{S}_0 = \left(\vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0 \vec{1}_p \right)^{-\frac{1}{2}} \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0,$$

which is the fourth row of [Table II](#).

Q.E.D.

PROOF of [Proposition 3](#) Note that the model implies:

$$\text{Cov}(\mathbf{y}, F) = \lambda_i \text{Var}(F)$$

Let $\Sigma := \text{Var}[\mathbf{y}] = \lambda \lambda' \text{Var}(F) + \Psi$, by the Sherman-Morrison formula,

$$\Sigma^{-1} = \Psi^{-1} - \frac{\Psi^{-1} \lambda \lambda' \text{Var}(F) \Psi^{-1}}{1 + \text{Var}(F) \lambda' \Psi^{-1} \lambda}.$$

The population version of the weighting vector A^{IC} as presented in [Table II](#), satisfies:

$$A^{\text{IC}} \propto_+ \vec{1}_p' S_0 \Sigma^{-1} S_0,$$

where $\propto_+$ denotes positive proportionality, and S_0 the diagonal matrix of control-group standard deviations equals:

$$\begin{aligned}
S_0 &= \text{Diag}\left(\sqrt{\lambda_1^2 \text{Var}_0(F) + \sigma_1^2}, \dots, \sqrt{\lambda_p^2 \text{Var}_0(F) + \sigma_p^2}\right) \Rightarrow \\
&\Rightarrow A^{\text{IC}} \propto_{+} \vec{1}'_p S_0 \Psi^{-1} S_0 - \text{Var}(F) \frac{\vec{1}'_p S_0 \Psi^{-1} \lambda}{1 + \text{Var}(F) \lambda' \Psi^{-1} \lambda} \lambda' \Psi^{-1} S_0 \Rightarrow \\
a_i &\propto_{+} \frac{\text{Var}_0(y_i)}{\sigma_i^2} - \frac{\vec{1}'_p S_0 \Psi^{-1} \lambda}{1 + \text{Var}(F) \lambda' \Psi^{-1} \lambda} \lambda_i \text{Var}(F) \frac{\sqrt{\text{Var}_0(y_i)}}{\sigma_i^2} = \\
&= \frac{\text{Var}_0(y_i)}{\sigma_i^2} \left(1 - \frac{\vec{1}'_p S_0 \Psi^{-1} \lambda}{1 + \text{Var}(F) \lambda' \Psi^{-1} \lambda} \frac{\text{Cov}(y, F)}{\sqrt{\text{Var}_0(y_i)}}\right) \Rightarrow \\
a_i &< 0 \Leftrightarrow \frac{\text{Cov}(y, F)}{\sqrt{\text{Var}_0(y_i)}} > \frac{1 + \text{Var}(F) \lambda' \Psi^{-1} \lambda}{\vec{1}'_p S_0 \Psi^{-1} \lambda} =: r.
\end{aligned}$$

Q.E.D.

PROPOSITION 8 – Asymptotic linearity of **SN weights:** Suppose $\{W_i\}_{i=1}^n$ are i.i.d., $D_i \in \{0, 1\}$, and $p_0 := P(D = 0) > 0$. For every component j used in at least one **SN** summary index, let

$$\mu_{j,0} := E_P[y_j \mid D = 0], \sigma_{j,0}^2 := \text{Var}_P[y_j \mid D = 0], \quad (21)$$

and assume $0 < \sigma_{j,0}^2 < \infty$ and $E_P[(y_j - \mu_{j,0})^4 \mid D = 0] < \infty$.

Let q non-standardized **SN** summary indices be constructed from nonempty sets $\mathcal{G}_\ell \subseteq \{1, \dots, p\}$, $\ell = 1, \dots, q$, where step 4 of [Procedure 1](#) is omitted. Let $A_n^{\text{SN}} = (a_{n,\ell,j})_{\ell=1,\dots,q;j=1,\dots,p}$ be the corresponding weighting matrix, with

$$a_{n,\ell,j} = \begin{cases} 0 & \text{if } j \notin \mathcal{G}_\ell \\ (|\mathcal{G}_\ell| \sqrt{\hat{\sigma}_{j,0,n}^2})^{-1} & \text{if } j \in \mathcal{G}_\ell \end{cases},$$

where $\hat{\sigma}_{j,0,n}^2$ is the control-group sample variance of component j . Then

$$A_n^{\text{SN}} = A^{\text{SN}} + \frac{1}{n} \sum_{i=1}^n \psi_P^{\text{SN}}(W_i) + o_p\left(\frac{1}{\sqrt{n}}\right),$$

where

$$a_{\ell,j} = \begin{cases} 0 & \text{if } j \notin \mathcal{G}_\ell \\ (|\mathcal{G}_\ell| \sigma_{j,0})^{-1} & \text{if } j \in \mathcal{G}_\ell \end{cases}$$

and the components of $\psi_P^{\text{SN}}(W_i)$ are

$$\psi_P^{\text{SN}}(W_i)_{\ell,j} = \begin{cases} 0 & \text{if } j \notin \mathcal{G}_\ell \\ -\frac{1}{2|\mathcal{G}_\ell|} \frac{1-D_i}{p_0} \frac{(y_{j,i} - \mu_{j,0})^2 - \sigma_{j,0}^2}{(\sigma_{j,0}^2)^{\frac{3}{2}}} & \text{if } j \in \mathcal{G}_\ell. \end{cases}$$

In particular, $E_P[\psi_P^{\text{SN}}(W)] = 0$ and $E_P[\|\psi_P^{\text{SN}}(W)\|^2] < \infty$.

PROOF Consider an arbitrary element (ℓ, j) of A_n^{SN} . If $j \notin \mathcal{G}_\ell$, then $a_{n,\ell,j} = a_{\ell,j} = 0$ and the result is immediate. Let $j \in \mathcal{G}_\ell$ and write $c_\ell := \frac{1}{|\mathcal{G}_\ell|}$.

Define the control-group empirical moments

$$\hat{\theta}_{1,n} := \left(\sum_{i=1}^n (1 - D_i) \right)^{-1} \sum_{i=1}^n (1 - D_i) y_{j,i},$$

$$\hat{\theta}_{2,n} := \left(\sum_{i=1}^n (1 - D_i) \right)^{-1} \sum_{i=1}^n (1 - D_i) y_{j,i}^2,$$

and their population analogues $\theta_1 := E_P[y_j \mid D = 0]$ and $\theta_2 := E_P[y_j^2 \mid D = 0]$. Then $\sigma_{j,0}^2 = \theta_2 - \theta_1^2$ and

$$a_{n,\ell,j} = f(\hat{\theta}_{1,n}, \hat{\theta}_{2,n}), \quad f(x_1, x_2) := c_\ell (x_2 - x_1^2)^{-\frac{1}{2}}.$$

By the standard expansion for conditional sample moments,

$$\begin{pmatrix} \hat{\theta}_{1,n} - \theta_1 \\ \hat{\theta}_{2,n} - \theta_2 \end{pmatrix} = \frac{1}{n} \sum_{i=1}^n \frac{1 - D_i}{p_0} \begin{pmatrix} y_{j,i} - \theta_1 \\ y_{j,i}^2 - \theta_2 \end{pmatrix} + o_p\left(\frac{1}{\sqrt{n}}\right).$$

The fourth-moment and positive-variance assumptions imply that f is continuously differentiable in a neighborhood of (θ_1, θ_2) with probability approaching one. A first-order Taylor expansion gives

$$a_{n,\ell,j} - a_{\ell,j} = (\nabla f(\theta_1, \theta_2))' \frac{1}{n} \sum_{i=1}^n \frac{1 - D_i}{p_0} \begin{pmatrix} y_{j,i} - \theta_1 \\ y_{j,i}^2 - \theta_2 \end{pmatrix} + o_p\left(\frac{1}{\sqrt{n}}\right).$$

Since

$$\nabla f(\theta_1, \theta_2) = c_\ell (\sigma_{j,0}^2)^{-\frac{3}{2}} \begin{pmatrix} \theta_1 \\ -\frac{1}{2} \end{pmatrix},$$

the summand equals

$$-\frac{1}{2|\mathcal{G}_\ell|} \frac{1 - D_i}{p_0} \frac{(y_{j,i} - \mu_{j,0})^2 - \sigma_{j,0}^2}{(\sigma_{j,0}^2)^{\frac{3}{2}}},$$

which is the displayed expression for $\psi_P^{\text{SN}}(W_i)_{\ell,j}$. Because the number of matrix entries is fixed, applying this argument element by element proves the matrix representation. The zero mean and finite second moment of $\psi_P^{\text{SN}}(W)$ follow from the displayed formula and the assumed fourth moments. Q.E.D.

PROOF of Proposition 4 By Proposition 1, $\hat{\beta}_n = \hat{A}_n \hat{\tau}_n$. The asymptotic linear representations in Assumptions 2 and 3 imply $\hat{\tau}_n - \tau = O_{p(\frac{1}{\sqrt{n}})}$ and $\hat{A}_n - A = O_{p(\frac{1}{\sqrt{n}})}$. Therefore,

$$\begin{aligned} \sqrt{n}(\hat{\beta}_n - \beta) &= \sqrt{n}(\hat{A}_n \hat{\tau}_n - A\tau) \\ &= A\sqrt{n}(\hat{\tau}_n - \tau) + \sqrt{n}(\hat{A}_n - A)\tau + \sqrt{n}(\hat{A}_n - A)(\hat{\tau}_n - \tau). \end{aligned}$$

The last term is $o_{p(1)}$ because it is $\sqrt{n}O_{p(\frac{1}{\sqrt{n}})}O_{p(\frac{1}{\sqrt{n}})}$. Substituting the two asymptotic linear representations gives

$$\sqrt{n}(\hat{\beta}_n - \beta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n (A \vartheta(W_i) + \psi(W_i) \tau) + o_{p(1)}.$$

The summand has mean zero and finite second moment by Assumptions 2 and 3. The multivariate Lindeberg-Levy central limit theorem therefore implies

$$\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} \mathcal{N}(0, \text{Var}(A \vartheta(W) + \psi(W) \tau)).$$

This is (14). Consistency follows from the same expansion, since $\hat{\beta}_n - \beta = O_{p(\frac{1}{\sqrt{n}})}$. Q.E.D.

ASSUMPTION 6: *Mean-square regularity.* Along the sequence in [Assumption 4](#),

$$E_n \left\| \sqrt{n}(\hat{\tau}_n - \tau_n) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \vartheta(W_i) \right\|^2 \rightarrow 0,$$

and

$$E_n \left\| \sqrt{n}(\hat{A}_n - A) \right\|^4 = O(1), E_n \left\| \sqrt{n}(\hat{\tau}_n - \tau_n) \right\|^4 = O(1).$$

PROOF of [Proposition 5](#) Let $\tau_n = \frac{\delta}{\sqrt{n}}$ and $\beta_n = A\tau_n$. By [Proposition 1](#), $\hat{\beta}_n = \hat{A}_n \hat{\tau}_n$. Under [3](#), $\hat{A}_n - A = O_p(\frac{1}{\sqrt{n}})$; under [4](#), $\hat{\tau}_n - \tau_n = O_p(\frac{1}{\sqrt{n}})$. Hence

$$\sqrt{n}(\hat{\beta}_n - \beta_n) = A\sqrt{n}(\hat{\tau}_n - \tau_n) + \sqrt{n}(\hat{A}_n - A)\tau_n + \sqrt{n}(\hat{A}_n - A)(\hat{\tau}_n - \tau_n).$$

For part (i), impose [Assumption 6](#) and define

$$X_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n A \vartheta(W_i), R_{\tau,n} := \sqrt{n}(\hat{\tau}_n - \tau_n) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \vartheta(W_i).$$

The first condition in [Assumption 6](#) gives $E_n \|R_{\tau,n}\|^2 \rightarrow 0$. Moreover, because $\tau_n = \frac{\delta}{\sqrt{n}}$,

$$E_n \left\| \sqrt{n}(\hat{A}_n - A)\tau_n \right\|^2 \leq \frac{\|\delta\|^2}{n} E_n \left\| \sqrt{n}(\hat{A}_n - A) \right\|^2 \rightarrow 0,$$

and Cauchy-Schwarz together with the fourth-moment bounds in [Assumption 6](#) gives

$$E_n \left\| \sqrt{n}(\hat{A}_n - A)(\hat{\tau}_n - \tau_n) \right\|^2 \leq \frac{1}{n} \left(E_n \left\| \sqrt{n}(\hat{A}_n - A) \right\|^4 E_n \left\| \sqrt{n}(\hat{\tau}_n - \tau_n) \right\|^4 \right)^{\frac{1}{2}} \rightarrow 0.$$

Thus,

$$\sqrt{n}(\hat{\beta}_n - \beta_n) = X_n + r_n, E_n r_n^2 \rightarrow 0.$$

By [Assumption 4](#), $E_n X_n = 0$ and $E_n X_n^2 = AVA'$. Therefore, by Cauchy-Schwarz,

$$nE_n \left[(\hat{\beta}_n - \beta_n)^2 \right] = E_n (X_n + r_n)^2 = AVA' + o(1),$$

which proves [\(18\)](#).

The same expansion in probability gives

$$\sqrt{n}(\hat{\beta}_n - \beta_n) = \frac{1}{\sqrt{n}} \sum_{i=1}^n A \vartheta(W_i) + o_{p(1)} \xrightarrow{d} \mathcal{N}(0, AVA').$$

Since $\sqrt{n}\beta_n = A\delta$,

$$\sqrt{n}\hat{\beta}_n = A\delta + \frac{1}{\sqrt{n}} \sum_{i=1}^n A \vartheta(W_i) + o_{p(1)}.$$

Under the standard-error consistency condition in part (ii),

$$T_n = \frac{\hat{\beta}_n}{\text{s.e.}(\hat{\beta}_n)} \xrightarrow{d} \mathcal{N}(\Delta, 1), \Delta = \frac{A\delta}{\sqrt{AVA'}}.$$

Therefore, for $z = z_{1-\alpha/2}$,

$$\pi(\Delta) = P(|Z + \Delta| > z) = \Phi(\Delta - z) + \Phi(-\Delta - z),$$

where $Z \sim \mathcal{N}(0, 1)$. This proves (19).

It remains to characterize the maximizing direction. The function π is even, and for $x \geq 0$ its derivative is $f(x - z) - f(x + z) \geq 0$, where f is the standard normal density. Hence maximizing local power is equivalent to maximizing $|\Delta|$. For any fixed value of $\delta'V^{-1}\delta$, the denominator in Δ is fixed, so it remains to maximize $|A\delta|$. By Cauchy-Schwarz,

$$|A\delta|^2 = \left| (V^{1/2}A')' (V^{-1/2}\delta) \right|^2 \leq (AVA')(\delta'V^{-1}\delta).$$

Equality holds if and only if $V^{-1/2}\delta \propto V^{1/2}A'$, equivalently $\delta \propto VA'$. Hence the directions proportional to VA' attain the maximal value of $|\Delta|$ and thus the maximal two-sided local power. Q.E.D.

PROOF of Corollary 3.1 Let $W_n := A_n \hat{S}_0$ denote the row vector reported in Table II; this is the row vector multiplying the standardized estimator $\hat{\tau}_{n, \text{st}}$. Since $\hat{\tau}_n = \hat{S}_0 \hat{\tau}_{n, \text{st}}$, the row vector multiplying the unstandardized estimator $\hat{\tau}_n$ is $A_n = W_n \hat{S}_0^{-1}$. The assumptions $\hat{\Sigma} \xrightarrow{p} \Sigma$ and $\hat{\Sigma}_0 \xrightarrow{p} \Sigma_0$, together with positive definiteness, imply by the continuous mapping theorem that $\hat{S}_0 \xrightarrow{p} S_0$, $\hat{S}_0^{-1} \xrightarrow{p} S_0^{-1}$, and $\hat{\Sigma}^{-1} \xrightarrow{p} \Sigma^{-1}$.

For **SN**, Table II gives $W_n = c_n \vec{1}_p'$ for a positive scalar c_n ; this scalar is p^{-1} without final standardization and another positive normalizing constant with final standardization. Thus

$$A^{\text{SN}} \propto \vec{1}_p' S_0^{-1}, (A^{\text{SN}})' \propto S_0^{-1} \vec{1}_p.$$

Applying [Proposition 5](#) gives

$$\delta_{\text{SN}}^* \propto V (A^{\text{SN}})' \propto V S_0^{-1} \vec{1}_p.$$

For **IC**, [Table II](#) gives

$$W_n = d_n \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1} \hat{S}_0$$

for a positive scalar d_n ; again, the scalar differs depending on whether the index is finally standardized. Hence

$$A_n = W_n \hat{S}_0^{-1} = d_n \vec{1}_p' \hat{S}_0 \hat{\Sigma}^{-1},$$

and therefore

$$A^{\text{IC}} \propto \vec{1}_p' S_0 \Sigma^{-1}, (A^{\text{IC}})' \propto \Sigma^{-1} S_0 \vec{1}_p,$$

using symmetry of S_0 and Σ . Applying [Proposition 5](#) gives

$$\delta_{\text{IC}}^* \propto V (A^{\text{IC}})' \propto V \Sigma^{-1} S_0 \vec{1}_p.$$

This proves the two expressions in [Corollary 3.1](#).

Q.E.D.

PROOF of [Proposition 6](#) Under [Assumption 5](#), the difference-in-means estimator satisfies

$$\sqrt{n}(\hat{\tau}_n - \tau_n) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left(\frac{D_i}{\nu} - \frac{1-D_i}{1-\nu} \right) \mathbf{u}_i + o_p(1).$$

Therefore the asymptotic variance matrix in [\(18\)](#) is

$$V_\tau = (\nu(1-\nu))^{-1} \mathcal{V}.$$

The local treatment effect contributes only $O(\frac{1}{n})$ to the covariance matrix of $\mathbf{y}_{i,n}$. Hence the full-sample and control-group covariance matrices both converge to $\mathcal{V}$, and $S_0 = \text{Diag}(\mathcal{V})^{\frac{1}{2}}$. Let $C := S_0^{-1} \mathcal{V} S_0^{-1}$.

Let W denote the limiting row vector that a procedure in [Table II](#) applies to the standardized estimator $\hat{\tau}_{n,\text{st}}$. Since $\hat{\tau}_{n,\text{st}} = S_0^{-1} \hat{\tau}_n$, the corresponding row vector on the unstandardized estimator is $A = W S_0^{-1}$. Thus [\(18\)](#) gives

$$\text{MSE}(W) = A V_\tau A' = (\nu(1-\nu))^{-1} W C W'.$$

For **SN** without final standardization, $W_{\text{SN}}^0 = p^{-1} \vec{1}_p'$, so

$$\text{MSE}_{\text{SN}}^0 = (\nu(1 - \nu))^{-1} \frac{\vec{1}_p' C \vec{1}_p}{p^2}.$$

With final standardization, $W_{\text{SN}}^1 = (\vec{1}_p' C \vec{1}_p)^{-\frac{1}{2}} \vec{1}_p'$, and hence

$$\text{MSE}_{\text{SN}}^1 = (\nu(1 - \nu))^{-1}.$$

For **IC** without final standardization, [Table II](#) gives $W_{\text{IC}}^0 = (\vec{1}_p' C^{-1} \vec{1}_p)^{-1} \vec{1}_p' C^{-1}$, so

$$\text{MSE}_{\text{IC}}^0 = (\nu(1 - \nu))^{-1} (\vec{1}_p' C^{-1} \vec{1}_p)^{-1}.$$

With final standardization, $W_{\text{IC}}^1 = (\vec{1}_p' C^{-1} \vec{1}_p)^{-\frac{1}{2}} \vec{1}_p' C^{-1}$, and therefore

$$\text{MSE}_{\text{IC}}^1 = (\nu(1 - \nu))^{-1}.$$

It remains to compare the two non-standardized procedures. Since C is positive definite,

$$p^2 = (\vec{1}_p' \vec{1}_p)^2 = \left((C^{\frac{1}{2}} \vec{1}_p)' (C^{-\frac{1}{2}} \vec{1}_p) \right)^2 \leq (\vec{1}_p' C \vec{1}_p) (\vec{1}_p' C^{-1} \vec{1}_p),$$

by Cauchy-Schwarz. This proves $\text{MSE}_{\text{IC}}^0 \leq \text{MSE}_{\text{SN}}^0$. Finally, C is a correlation matrix, so $C_{j,k} \leq 1$ for all j, k , and therefore $\vec{1}_p' C \vec{1}_p \leq p^2$. This proves $\text{MSE}_{\text{SN}}^0 \leq \text{MSE}_{\text{SN}}^1 = \text{MSE}_{\text{IC}}^1$. Q.E.D.

COROLLARY 1.1 — Local power of Bonferroni-corrected component tests: Let $T_{j,n} := \frac{\hat{\tau}_{j,n}}{\text{s.e.}(\hat{\tau}_{j,n})}$ be the component-level t -statistic for component j , where $\sqrt{n} \text{s.e.}(\hat{\tau}_{j,n})$ is consistent for $\sqrt{V_{j,j}}$. Let $\mathbf{T}_n := (T_{1,n}, \dots, T_{p,n})'$ and define $V_D := \text{Diag}(V_{1,1}, \dots, V_{p,p})$. Under [Assumption 4](#),

$$\mathbf{T}_n \xrightarrow{d} \mathcal{N}(V_D^{-1/2} \boldsymbol{\delta}, V_D^{-1/2} V V_D^{-1/2}). \quad (22)$$

Therefore, for the two-sided Bonferroni rule that rejects $H_0 : \boldsymbol{\tau} = \mathbf{0}$ whenever $\max_{1 \leq j \leq p} |T_{j,n}| > c_p$, with $c_p := z_{1-\alpha/(2p)}$, the asymptotic local power equals

$$\pi_{B(\boldsymbol{\delta}, V)} := P\left(\max_{1 \leq j \leq p} |Z_j| > c_p\right),$$

where $\mathbf{Z}$ has the distribution in [\(22\)](#).

PROOF of Corollary 1.1 Under [Assumption 4](#),

$$\sqrt{n}(\hat{\tau}_n - \tau_n) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \vartheta(W_i) + o_{p(1)},$$

and $\tau_n = \frac{\delta}{\sqrt{n}}$. Hence the central limit theorem gives

$$\sqrt{n}\hat{\tau}_n = \delta + \frac{1}{\sqrt{n}} \sum_{i=1}^n \vartheta(W_i) + o_{p(1)} \xrightarrow{d} \mathcal{N}(\delta, V).$$

Let $V_D := \text{Diag}(V_{1,1}, \dots, V_{p,p})$. By the standard-error consistency assumed in [Corollary 1.1](#),

$$\text{Diag}(\sqrt{n} \text{s.e.}(\hat{\tau}_{1,n}), \dots, \sqrt{n} \text{s.e.}(\hat{\tau}_{p,n})) \xrightarrow{p} V_D^{\frac{1}{2}}.$$

Slutsky's theorem therefore implies

$$T_n = \text{Diag}(\sqrt{n} \text{s.e.}(\hat{\tau}_{1,n}), \dots, \sqrt{n} \text{s.e.}(\hat{\tau}_{p,n}))^{-1} \sqrt{n}\hat{\tau}_n \xrightarrow{d} \mathcal{N}\left(V_D^{-\frac{1}{2}}\delta, V_D^{-\frac{1}{2}}V V_D^{-\frac{1}{2}}\right),$$

which is [\(22\)](#).

Let $B := \{z \in \mathbb{R}^p : \max_{1 \leq j \leq p} |z_j| > c_p\}$. The boundary of B is contained in the finite union of hyperplanes $\{z : z_j = c_p\}$ and $\{z : z_j = -c_p\}$, $j = 1, \dots, p$. The limiting normal distribution is nondegenerate, so this boundary has probability zero. The portmanteau theorem then gives

$$\lim_{n \rightarrow \infty} P_n\left(\max_{1 \leq j \leq p} |T_{j,n}| > c_p\right) = P\left(\max_{1 \leq j \leq p} |Z_j| > c_p\right),$$

where Z has the distribution in [\(22\)](#). This is the displayed expression for $\pi_{B(\delta, V)}$. *Q.E.D.*

B. OTHER SUMMARY INDICES

The results in the main text do not depend on the literal steps in [Procedure 1](#) or [2](#). They use the fact that, after the data-dependent weights are fixed, the final scalar index can be written as

$$s_{i,n} = A_n \mathbf{y}_i + b_n.$$

For any index with this representation, the same argument as in [Proposition 1](#) gives $\hat{\beta}_n = A_n \hat{\tau}_n$. The large-sample results then apply whenever A_n satisfies the expansion in [Assumption 3](#).

This covers the following common variants.

- (i) *Multiple and overlapping domains.* Researchers may partition outcomes into several domains, or allow domains to overlap, and construct one index per domain, as in [Casey, Glennerster, and Miguel \(2012\)](#). The scalar results apply to each domain separately.

Equivalently, if the q indices are treated jointly, A_n becomes a $q \times p$ matrix whose rows give the weights of each index on the original outcomes.

- (ii) *Nested indices*. If an index is constructed from sub-indices, as in [Haushofer and Shapiro \(2016\)](#) and [Christensen et al. \(2021\)](#), the final index is still affine in the original outcomes whenever each construction step is affine. The relevant weights are the final weights on the original outcomes, not only the weights assigned to the intermediate sub-indices.
- (iii) *PCA indices*. If the first principal component is used as the index, as in [Filmer and Pritchett \(2001\)](#), the index is linear in the standardized outcomes conditional on the estimated loading vector. The results therefore apply under regularity conditions that make this loading vector asymptotically linear, such as a separated first eigenvalue and a fixed sign normalization.
- (iv) *Economically weighted indices*. If outcomes are weighted by external costs or benefits, as in [Bhatt et al. \(2024\)](#), the representation above holds. If the weights are fixed external constants, the generated-weight term is zero; if the weights are estimated, the same results apply after accounting for their influence function.

This discussion assumes that, for each constructed index, the same row vector A_n applies to every observation. Missing-value imputations that create observation-specific weights are treated separately in Appendix C.

C. MISSING DATA

When a component of $\mathbf{y}$ is missing for some observation, a common practice is to construct the summary index only from the components observed for that observation, rather than requiring complete data or dropping the observation entirely. Under such an **available-components rule**, each observed component absorbs the weight that would otherwise have gone to the missing components for that observation, so a component’s realized weight depends on which other components happen to be missing alongside it. [Anderson \(2008\)](#) motivates the missing-value adjustment used in the IC procedure on this basis, stating that it “uses all of the available data, but it weights outcomes with fewer missing values more heavily.” This appendix formalizes the rule and shows that this intuition does not generally hold: whether a component ends up with more or less weight on average is not simply governed by how often it is missing.

The reason this requires separate treatment is that the decomposition in [Proposition 1](#) requires a single row vector A_n such that $s_{i,n} = A_n \mathbf{y}_i + b_n$ for every observation, which is what lets the summary-index-level effect be read off as the fixed linear combination $A_n \hat{\boldsymbol{\tau}}_n$

of component-level effects. An available-components rule generally violates this, because the weight vector it induces varies with each observation's missingness pattern.

Formally, let $m_{i,j} = 1$ if component j is observed for observation i , and let $c_{j,n}$ denote the weight component j would receive in the absence of missing data. The available-components rule renormalizes these base weights over only the components observed for i :

$$s_{i,n}^{\text{miss}} = \frac{\sum_{j=1}^p m_{i,j} c_{j,n} \mathbf{y}_{i,j}}{\sum_{k=1}^p m_{i,k} c_{k,n}}.$$

Equivalently, $s_{i,n}^{\text{miss}} = A_{i,n}^{\text{miss}} \mathbf{y}_i$, with observation-specific weights

$$A_{i,n,j}^{\text{miss}} = \frac{m_{i,j} c_{j,n}}{\sum_{k=1}^p m_{i,k} c_{k,n}}.$$

that depend on i only through its missingness pattern $(m_{i,1}, \dots, m_{i,p})$, and so generally differ across observations. Averaging these observation-specific weights over the sample gives

$$\hat{\beta}_n^{\text{miss}} = \frac{1}{n} \sum_{i=1}^n A_{i,n}^{\text{miss}} \mathbf{y}_i \omega(\tilde{x}_i, \hat{\nu}_n),$$

which is not generally equal to $A_n \hat{\tau}_n$ for any single row vector A_n applied to the component estimates in [Proposition 1](#): no fixed set of index weights rationalizes $\hat{\beta}_n^{\text{miss}}$ as a linear combination of the component treatment effects. This is an algebraic issue separate from the missing-data assumptions needed for point identification – it arises even when missingness is unrelated to potential outcomes.

The following table quantifies why the “more weight to less-missing outcomes” intuition does not generally hold, reporting the weights induced by the equal-weight version of the available-components rule, $c_{j,n} = 1$, for three outcomes.

TABLE X
EFFECTIVE WEIGHTS UNDER AN AVAILABLE-COMPONENTS RULE

Obs.	Observed components	$j = 1$	$j = 2$	$j = 3$
1	2	0	1	0
2	1, 3	$\frac{1}{2}$	0	$\frac{1}{2}$
3	1, 2	$\frac{1}{2}$	$\frac{1}{2}$	0
4	1, 3	$\frac{1}{2}$	0	$\frac{1}{2}$
5	2	0	1	0
6	1, 3	$\frac{1}{2}$	0	$\frac{1}{2}$
7	1, 2	$\frac{1}{2}$	$\frac{1}{2}$	0
8	1, 2, 3	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
Missing observations		2	3	4
Average effective weight		$\frac{17}{48}$	$\frac{20}{48}$	$\frac{11}{48}$

Notes: Columns $j = 1, 2, 3$ report the weights $A_{i,n,j}^{\text{miss}}$ in $s_{i,n}^{\text{miss}} = \sum_{j=1}^3 A_{i,n,j}^{\text{miss}} y_{i,j}$ when all observed components receive equal base weight. The final two rows report the number of missing observations and the average effective weight for each outcome.

Outcome 1 has the fewest missing observations, but its average effective weight is smaller than the average effective weight assigned to outcome 2. The reason is that the weight assigned to an observed component depends on which other components are observed for the same individual. In the IC procedure, covariance weighting adds another source of variation, so the effective weights are even less tightly tied to marginal missingness rates. A cleaner approach is to estimate the component effects and the index weights under explicit assumptions on the missing-data mechanism, and then form $A_n \hat{\tau}_n$ directly.